

Enhancing Fundamental Reasoning with Reinforcement Learning: A Theoretical Framework for Reinforcement-Guided Reasoning (RGR)

Abstract—Reinforcement Learning (RL) has achieved remarkable success in complex decision-making tasks, yet it often struggles with abstract, multi-step reasoning and generalization. Traditional RL agents excel at learning policies from direct environmental interaction but lack the underlying logical or causal understanding characteristic of human cognition. In this paper, we introduce a theoretical framework called Reinforcement-Guided Reasoning (RGR). RGR reframes the problem of reasoning as a sequential decision-making process where the actions are not physical manipulations of an environment, but cognitive operations within a formal reasoning space. We propose a meta-learning architecture where an RL agent learns an optimal policy for selecting a sequence of reasoning primitives (e.g., applying a logical rule, querying a knowledge base, or forming a hypothesis) to solve complex problems efficiently. By learning to navigate the space of possible reasoning paths, the RGR agent can discover more efficient and generalizable problem-solving strategies than those offered by brute-force search or pre-programmed heuristics. This paper outlines the formal components of the RGR framework, analyzes its theoretical implications for generalization and interpretability, and discusses key challenges and avenues for future research.

Index Terms—Reinforcement Learning, Fundamental Reasoning, Meta-Learning, Cognitive Architectures, Neuro-Symbolic AI, Decision Making.

I. INTRODUCTION

Deep Reinforcement Learning (RL) has demonstrated superhuman performance in domains with well-defined rules and objectives, such as the games of Go [1]. These successes are largely attributed to the ability of deep neural networks to learn complex value functions and policies from vast amounts of experience. However, a significant gap remains between the pattern-matching prowess of current RL agents and the flexible, abstract reasoning capabilities of humans. An agent trained to escape one specific maze may fail completely in a new maze that is topologically distinct, even if the underlying principles of navigation remain the same. This limitation stems from the fact that the agent learns a policy over low-level actions (e.g., ‘move-up’, ‘move-down’) rather than a policy over high-level reasoning strategies (e.g., ‘identify the exit’, ‘follow the left wall’).

Fundamental reasoning encompasses a range of cognitive abilities, including logic, planning, causal inference, and analogy. These are the building blocks that allow humans to solve novel problems, adapt to new information, and make sense of the world. While symbolic AI (GOFAI) has long focused on these areas, it often suffers from brittleness and a lack of scalability [2].

This paper proposes a hybrid approach, **Reinforcement-Guided Reasoning (RGR)**, that leverages the adaptive, reward-driven nature of RL to navigate and optimize formal reasoning processes. The core idea is to treat a problem-solving task not as a single, monolithic policy-learning challenge, but as a meta-level MDP where the agent learns *how to reason*. The RGR framework is not intended to replace symbolic reasoners or large language models but to provide a principled mechanism for guiding their application in a goal-directed manner.

Our contribution is a formal theoretical framework that:

- 1) Defines a cognitive MDP for reasoning tasks.
- 2) Proposes an architecture where an RL agent selects from a set of discrete reasoning primitives.
- 3) Theorizes how this approach can lead to enhanced generalization, interpretability, and efficiency in complex problem-solving.

II. BACKGROUND AND RELATED WORK

A. Reinforcement Learning

RL is typically formalized as a Markov Decision Process (MDP), defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}$ is the state transition function, $\mathcal{R}$ is the reward function, and γ is the discount factor. The goal of an RL agent is to learn a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximizes the expected cumulative discounted reward.

B. Reasoning in Artificial Intelligence

AI research on reasoning has historically been divided. **Symbolic AI** uses formal logic and manipulation of symbols to perform tasks like theorem proving and planning [2]. These systems are highly interpretable but can be brittle. **Connectionist AI**, particularly deep learning, excels at learning from raw perceptual data but its reasoning processes are often opaque and struggle with systematic generalization [3].

Recent interest in **Neuro-Symbolic AI** seeks to bridge this gap by integrating neural networks with symbolic reasoning systems [4]. The RGR framework aligns with this paradigm, using a neural-based RL agent to direct a potentially symbolic reasoning engine.

III. THE REINFORCEMENT-GUIDED REASONING (RGR) FRAMEWORK

We propose to model the process of reasoning as a cognitive MDP. In this formulation, the RL agent does not interact with

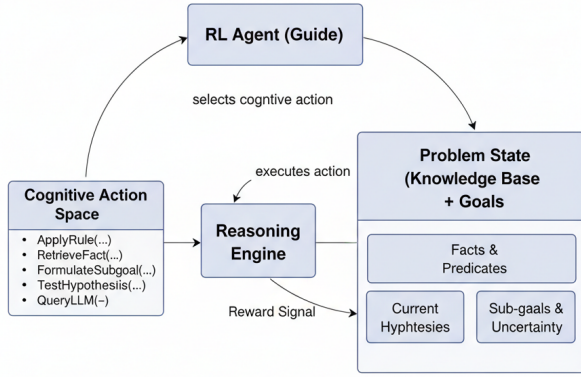


Fig. 1: Conceptual architecture of the RGR framework. The RL Agent (Guide) selects a cognitive action, which the Reasoning Engine executes to update the Problem State. A reward signal informs the agent’s policy.

a physical world but with a “reasoning environment” defined by the problem itself.

A. Formal Definition

The RGR framework can be described by a cognitive MDP, $M_c = (\mathcal{S}_c, \mathcal{A}_c, \mathcal{P}_c, \mathcal{R}_c, \gamma)$:

- **Cognitive State Space ($\mathcal{S}_c$):** A state $s_c \in \mathcal{S}_c$ represents the current state of knowledge about the problem. This can be represented as a set of known facts, logical predicates, current hypotheses, sub-goals, and a representation of the remaining uncertainty.
- **Cognitive Action Space ($\mathcal{A}_c$):** The action space consists of a discrete set of **reasoning primitives**. These are fundamental cognitive operations. Examples include:
 - `ApplyRule(rule_id, premises)`
 - `RetrieveFact(query)`
 - `FormulateSubgoal(description)`
 - `TestHypothesis(hypothesis)`
 - `QueryLLM(prompt)`
- **Cognitive Transition Function ($\mathcal{P}_c$):** The transition function $s'_c = f(s_c, a_c)$ is deterministic and executed by a **Reasoning Engine**. Given the current state s_c and the chosen cognitive action a_c , the engine computes the next reasoning state s'_c .
- **Cognitive Reward Function ($\mathcal{R}_c$):** The design of the reward function is critical. The agent must be incentivized to find correct solutions efficiently. The total reward at step t can be formulated as $R_t = r_{\text{step}} + r_{\text{intrinsic}} + R_{\text{terminal}}$.
 - **Terminal Reward:** A large positive reward for reaching a terminal state that correctly solves the problem.
 - **Intrinsic/Shaped Reward:** Dense, intermediate rewards for actions that reduce problem complexity, prove a useful lemma, or satisfy a sub-goal. A small

negative cost can be applied per step to encourage efficiency.

The RL agent, typically a deep Q-network or a policy gradient model, observes the current reasoning state s_c and learns a policy $\pi(a_c|s_c)$ to select the cognitive action that maximizes the expected future reward.

IV. THEORETICAL IMPLICATIONS AND ANALYSIS

A. Enhanced Generalization

By learning a policy over abstract reasoning primitives, the agent can generalize its problem-solving strategies. An agent that learns the optimal sequence of logical deductions to solve a specific Sudoku puzzle has learned a *strategy* for solving Sudoku, which can then be applied to any valid puzzle.

B. Interpretability

A significant drawback of deep learning models is their lack of transparency. The RGR framework provides a natural solution. The output is not just the final answer, but the entire sequence of cognitive actions selected by the RL agent, forming a human-readable trace of the model’s “thought process.”

C. Search Space Pruning

Many reasoning problems can be framed as a search through a vast combinatorial space. The RGR agent learns to navigate this search space intelligently. The learned policy $\pi(a_c|s_c)$ implicitly represents a heuristic for which reasoning paths are most promising, effectively pruning the search tree.

V. FUTURE WORK AND OPEN CHALLENGES

The RGR framework is theoretical and its practical implementation presents several challenges:

- 1) **Design of Primitive Sets:** The effectiveness of RGR is highly dependent on the choice of the cognitive action space $\mathcal{A}_c$. Creating a set of primitives that is both comprehensive and compact is a major design challenge.
- 2) **State Representation:** Representing the reasoning state s_c in a way that is suitable for a neural network is non-trivial. Graph-based representations may be a promising direction.
- 3) **Reward Shaping:** Designing a reward function that encourages complex, multi-step reasoning without being overly prescriptive is difficult.
- 4) **Scalability:** The cognitive state and action spaces can become extremely large for real-world problems. Developing RL algorithms that can handle this combinatorial complexity is a key area for future research.

VI. CONCLUSION

We have introduced Reinforcement-Guided Reasoning (RGR), a theoretical framework that combines the learning capabilities of reinforcement learning with the structured processes of formal reasoning. By training an RL agent to select a sequence of cognitive actions, RGR has the potential to create AI systems that are more generalizable, interpretable, and efficient at solving complex problems. This framework

provides a principled approach for unifying learning and reasoning, moving beyond pure pattern recognition towards a more robust and human-like form of artificial intelligence.

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